

Name: _____

SIMPLIFYING EXPRESSIONS, (COLLECTING LIKE TERMS)

Date: _____

LO: I can collect like terms to simplify algebraic expressions.

What are like terms?

Like terms have the **SAME** letter part (same variable raised to the same power). We can add or **subtract** their numbers.

Unlike terms have different letters or different powers. We cannot add or subtract them.

Examples of like terms

✓	$3x$ and $5x$	like terms
✓	$-2a$ and $7a$	like terms
✓	$4y^2$ and $-y^2$	like terms
✗	$3x$ and $3y$	not like terms
✗	$2a$ and $2a^2$	not like terms

Key idea

Collect the like terms together, then add or **subtract** the numbers.

$$\textcircled{3x} + \textcircled{5x} = \textcircled{8x}$$

$3 + 5 = 8$, keep the x

MODELLED EXAMPLES

We can simplify expressions by collecting like terms.

Example 1 – all positive

Simplify $3x + 5x$

$$\begin{array}{ccc} 3x & + & 5x \\ \swarrow & & \searrow \\ & 8x & \end{array}$$

$$= 8x$$

Add the numbers, keep the letter.

Example 2 – all negative

Simplify $-6a - 2a$

$$\begin{array}{ccc} -6a & - & 2a \\ \swarrow & & \searrow \\ & -8a & \end{array}$$

$$= -8a$$

Add the numbers ($-6 + -2 = -8$), keep the letter.

Example 3 – mixed signs

Simplify $7x - 4x$

$$\begin{array}{ccc} 7x & - & 4x \\ \swarrow & & \searrow \\ & 3x & \end{array}$$

$$= 3x$$

Subtract the numbers ($7 - 4 = 3$), keep the letter.

Example 4 – with unlike terms

Simplify $4a + 2b + 3a - b$

$$(4a + 3a) + (2b - b)$$

$$7a + b$$

$$= 7a + b$$

Collect like terms. Keep different letters separate.

YOUR TURN – PRACTICE (deliberate practice)

1. Identify the like terms

Circle the like terms in each expression.

- a) $3x + 5x$
- b) $4a - 2b + 7a$
- c) $2y + 3x + y$
- d) $5m^2 + 2m + 3m^2$

2. Collect like terms (all positive)

Simplify each expression.

- a) $2x + 6x$
- b) $4a + 5a + a$
- c) $3y + 7y + 2y$
- d) $8b + b$

3. Collect like terms (all negative)

Simplify each expression.

- a) $-3x - 5x$
- b) $-2a - 7a$
- c) $-y - 4y - 3y$
- d) $-6b - b$

4. Collect like terms (mixed signs)

Simplify each expression.

- a) $6x - 2x$
- b) $9a - 4a$
- c) $7y - 3y + 2y$
- d) $5b - 2b + b$

5. Collect like terms (mixed expressions)

Simplify each expression.

- a) $3x + 2x + 4$
- b) $5a - 2a + 7$
- c) $4p + 3q + p$
- d) $6m - m + 2m - 5$
- e) $2x + 3y + 4x - y$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick (✓) the ones that are correct.

If it is wrong, write the correct answer.

a) $3x + 5y = 8xy$ ☐

Correct answer: _____

b) $4a + 2a = 6a$ ☐

Correct answer: _____

c) $7x - 2x = 5$ ☐

Correct answer: _____

d) $2y + y - 5 = 3y - 5$ ☐

Correct answer: _____

e) $3p + 2q + p = 5pq$ ☐

Correct answer: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A ticket for a concert costs £3x for an adult and £2x for a child.

A family buys 2 adult tickets and 3 child tickets.

Write and simplify an expression for the total cost.

Expression: _____

Simplified: _____



b) A gardener plants x rose bushes, 2x lavender plants and 3x tulips. The next week, she plants 4 more rose bushes.

Write and simplify an expression for the total number of plants she has now.

Expression: _____

Simplified: _____



TOP TIPS



Look for the letters. Only combine the terms with the same letter (same power). Then add or subtract the numbers.



CHECK YOUR WORK

- ✓ Have I combined only like terms?
- ✓ Did I add or subtract the numbers correctly?
- ✓ Did I keep the letters?

WELL DONE!